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Final Report — Project-Specific Content

Formula 1 Race Intelligence Analysis Using Tableau

Dataset: F1 Racing Project Dataset(1).xlsx | Prepared 27 October 2026

1. INTRODUCTION — Formula 1 Race Intelligence Analysis Using Tableau turns the supplied workbook into an interactive analytical experience for comparing drivers, constructors, tyre usage, podiums and reliability patterns.

1.1 Project overview and purpose: The workbook contains five sheets: race_results, circuits, championship_standings, lap_intelligence and qualifying_results. The race_results sheet has 6,464 data rows. The aim is to make performance trends easier to inspect while keeping definitions and limitations transparent.

2. IDEATION — Primary users are team strategists, broadcast analysts and technical/reliability analysts. Their core need is to compare performance without manually scanning separate sheets. Prioritized ideas are a driver points leaderboard, constructor season trajectory, tyre analysis, podium distribution and DNF reliability analysis.

3. REQUIREMENT ANALYSIS — Functional requirements: connect and validate source sheets; create documented KPIs; provide season/team/driver/circuit filters; display charts and tooltips; assemble a story; verify results. Non-functional requirements: usability, consistent metric definitions, reproducibility, reasonable performance and safe handling of connection details.

3.1 Data flow: Excel workbook → validation and relationships → calculated fields → worksheets → dashboards → Tableau Story → optional publication. The source workbook remains unchanged.

4. PROJECT DESIGN — Use Tableau Desktop or Tableau Public. Relate race_results to circuits through circuit_id where appropriate, and connect season/driver-level data only after checking cardinality. Use relationships or pre-aggregated tables to avoid multiplying rows in one-to-many joins.

5. PLANNING — Suggested milestones: (1) profile workbook and validate data; (2) create driver and constructor worksheets; (3) build tyre, podium and reliability views; (4) assemble dashboards and story; (5) validate totals, rehearse and publish if approved. Team names, actual effort and publication links must be filled by the project team.

6. TESTING — Validate row counts and key uniqueness; compare Tableau aggregations against Excel; test each filter independently and in combination; check null handling, labels, tooltips and cross-sheet relationships. Record actual load/filter times only after executing tests.

7. RESULTS — Do not state visual findings until worksheets have been created and validated. Candidate analyses include driver points, constructor points by season, tyre compound frequency, tyre age vs lap time, podiums by driver/team, DNF by team/season/reason, circuit-type and wet-race comparisons, grid-to-finish position gains, and pit-stop/safety-car context.

8. ADVANTAGES AND DISADVANTAGES — Advantages: interactive comparisons, clear story flow, multiple levels of analysis and reusable filters. Limitations: source data may contain missing values; the workbook does not include driver nationality; results are descriptive rather than causal; event-level metrics require correct race_id grain.

9. CONCLUSION — A well-validated Tableau solution can help users explore Formula 1 performance and reliability questions efficiently. Trustworthiness depends on documented calculations, correct data relationships and avoiding unsupported claims.

10. FUTURE SCOPE — Add a sourced nationality mapping; compare with authoritative official timing data; automate refresh; add more seasons; develop reusable data-quality checks; optionally move to a warehouse such as Snowflake if the project requires scheduled refresh and larger-scale processing.

11. APPENDIX — Source workbook: F1 Racing Project Dataset(1).xlsx. Sheets: circuits (29 rows), championship_standings (311), lap_intelligence (3,814), qualifying_results (3,130), race_results (6,464). Observed race_results coverage: 2010–2026; 334 distinct race_id values; 19 driver labels; 10 team labels; 21 country labels.

Data caveats: 6,068 of 6,464 race_results rows have a null dnf_reason value; the dnf flag is present and has 396 true/1 records. No driver-nationality column is present. Counts of race events should use COUNTD(race_id) only after confirming the ID's event-level meaning; row counts represent driver-result records, not Grand Prix count.